

Midterm Exam 3 Retake Solutions & Rubric

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Suggested Point Distribution:

Undergraduate Total: 80 pts / Graduate Total: 100 pts

- **Exercise 1:** 20 points (5 pts for PFD, 5 pts per series region)
- **Exercise 2:** 20 points (5 pts for identifying poles, 10 pts for residues, 5 pts for final integral)
- **Exercise 3:** 20 points (5 pts for UHP roots, 10 pts for residues, 5 pts for final sum/evaluation)
- **Exercise 4:** 20 points (10 pts for z -substitution/expansion, 10 pts for evaluating the residue)
- **Exercise 5 (Grad):** 20 points (5 pts for poles/residues, 15 pts for sum setup and limit derivation)

Exercise 1

Consider the rational function: $f(z) = \frac{3}{(z-1)(z-4)}$

Solution: (a) **Partial Fraction Decomposition:**

$$\frac{3}{(z-1)(z-4)} = \frac{A}{z-1} + \frac{B}{z-4} \implies A(z-4) + B(z-1) = 3$$

Let $z = 1 \implies A(-3) = 3 \implies A = -1$.

Let $z = 4 \implies B(3) = 3 \implies B = 1$.

$$f(z) = -\frac{1}{z-1} + \frac{1}{z-4}$$

(b) **Series Expansions:**

i. **Region $|z| < 1$:** Both ratios $|z/1| < 1$ and $|z/4| < 1$.

$$-\frac{1}{z-1} = \frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$$

$$\frac{1}{z-4} = -\frac{1}{4(1-z/4)} = -\frac{1}{4} \sum_{n=0}^{\infty} \left(\frac{z}{4}\right)^n = -\sum_{n=0}^{\infty} \frac{z^n}{4^{n+1}}$$

$$f(z) = \sum_{n=0}^{\infty} \left(1 - \frac{1}{4^{n+1}}\right) z^n$$

ii. **Region $1 < |z| < 4$:** Here $|1/z| < 1$ and $|z/4| < 1$.

$$-\frac{1}{z-1} = -\frac{1}{z(1-1/z)} = -\frac{1}{z} \sum_{n=0}^{\infty} \left(\frac{1}{z}\right)^n = -\sum_{n=0}^{\infty} \frac{1}{z^{n+1}}$$

The second term remains the same as in region (i).

$$f(z) = -\sum_{n=0}^{\infty} \frac{1}{z^{n+1}} - \sum_{n=0}^{\infty} \frac{z^n}{4^{n+1}}$$

iii. Region $|z| > 4$: Both ratios $|1/z| < 1$ and $|4/z| < 1$.

$$\frac{1}{z-4} = \frac{1}{z(1-4/z)} = \frac{1}{z} \sum_{n=0}^{\infty} \left(\frac{4}{z}\right)^n = \sum_{n=0}^{\infty} \frac{4^n}{z^{n+1}}$$

The first term remains the same as in region (ii).

$$f(z) = \sum_{n=0}^{\infty} \frac{4^n - 1}{z^{n+1}}$$

Exercise 2

Evaluate $\oint_{|z|=3} \frac{z^2 - 3z}{(z+2i)^2(z-1)} dz$.

Solution: The integrand has two singularities inside the circle $|z| = 3$: a simple pole at $z = 1$ and a pole of order 2 at $z = -2i$.

Residue at $z = 1$:

$$\text{Res}(f, 1) = \left. \frac{z^2 - 3z}{(z+2i)^2} \right|_{z=1} = \frac{1-3}{(1+2i)^2} = \frac{-2}{1+4i-4} = \frac{-2}{-3+4i}$$

Multiply by the conjugate to clean up:

$$\frac{-2(-3-4i)}{(-3)^2 + 4^2} = \frac{6+8i}{25}$$

Residue at $z = -2i$: Use the derivative limit formula for $m = 2$:

$$\text{Res}(f, -2i) = \lim_{z \rightarrow -2i} \frac{d}{dz} \left[\frac{z^2 - 3z}{z-1} \right]$$

By the quotient rule:

$$\frac{d}{dz} \left[\frac{z^2 - 3z}{z-1} \right] = \frac{(2z-3)(z-1) - (z^2-3z)(1)}{(z-1)^2} = \frac{z^2 - 2z + 3}{(z-1)^2}$$

Evaluate at $z = -2i$:

$$\text{Numerator: } (-2i)^2 - 2(-2i) + 3 = -4 + 4i + 3 = -1 + 4i$$

$$\text{Denominator: } (-2i-1)^2 = (-1)^2 + 2(-1)(-2i) + (-2i)^2 = 1 + 4i - 4 = -3 + 4i$$

So the residue is:

$$\frac{-1+4i}{-3+4i} = \frac{(-1+4i)(-3-4i)}{25} = \frac{3+4i-12i+16}{25} = \frac{19-8i}{25}$$

Total Integral: By the Residue Theorem, we sum the residues and multiply by $2\pi i$:

$$\sum \text{Res} = \frac{6+8i}{25} + \frac{19-8i}{25} = \frac{25}{25} = 1$$

$$\oint_{|z|=3} f(z) dz = 2\pi i(1) = 2\pi i$$

Exercise 3

Evaluate the real integral: $\int_{-\infty}^{\infty} \frac{1}{x^4 + 81} dx$.

Solution: Let $f(z) = \frac{1}{z^4 + 81}$. We integrate over a large semicircular contour C_R in the upper half-plane. Solve $z^4 = -81 = 81e^{i\pi}$. The two roots in the UHP are:

$$z_0 = 3e^{i\pi/4} = \frac{3\sqrt{2}}{2}(1+i) \quad \text{and} \quad z_1 = 3e^{i3\pi/4} = \frac{3\sqrt{2}}{2}(-1+i)$$

Note that $z_0 + z_1 = 3\sqrt{2}i$.

Using the $P(z)/Q'(z)$ rule for the residues, $Q'(z) = 4z^3$. Since $z_k^4 = -81$, we have $z_k^3 = -81/z_k$.

$$\text{Res}(f, z_k) = \frac{1}{4z_k^3} = \frac{z_k}{4(-81)} = -\frac{z_k}{324}$$

Sum of the residues:

$$\text{Res}(f, z_0) + \text{Res}(f, z_1) = -\frac{z_0 + z_1}{324} = -\frac{3\sqrt{2}i}{324} = -i\frac{\sqrt{2}}{108}$$

By the Residue Theorem, the integral evaluates to:

$$\oint f(z) dz = 2\pi i \left(-i\frac{\sqrt{2}}{108} \right) = \frac{2\pi\sqrt{2}}{108} = \frac{\pi\sqrt{2}}{54}$$

The integral along the arc vanishes as $R \rightarrow \infty$ since the denominator is of 2 degrees higher than the numerator, making this the final real answer.

Exercise 4

Evaluate the definite integral: $\int_0^{2\pi} \cos^4(\theta) d\theta$.

Solution: Use the standard substitution $z = e^{i\theta} \implies d\theta = \frac{dz}{iz}$ and $\cos \theta = \frac{z+1/z}{2}$. The integral transforms into a contour integral over the unit circle $C: |z|=1$:

$$\int_0^{2\pi} \cos^4(\theta) d\theta = \oint_{|z|=1} \left(\frac{z+1/z}{2} \right)^4 \frac{dz}{iz} = \frac{1}{16i} \oint_{|z|=1} \left(z + \frac{1}{z} \right)^4 \frac{dz}{z}$$

Expand the binomial $(z + z^{-1})^4$:

$$(z + z^{-1})^4 = z^4 + 4z^2(1) + 6(z^2)(z^{-2}) + 4(1)(z^{-2}) + z^{-4} = z^4 + 4z^2 + 6 + 4z^{-2} + z^{-4}$$

Substitute this back into the integral:

$$\frac{1}{16i} \oint_{|z|=1} \left(z^3 + 4z + \frac{6}{z} + \frac{4}{z^3} + \frac{1}{z^5} \right) dz$$

The only term with a non-zero residue inside $|z| = 1$ is the $1/z$ term, which has a coefficient of 6.

$$\text{Res}(f, 0) = 6$$

Applying the Residue Theorem:

$$\int_0^{2\pi} \cos^4(\theta) d\theta = \frac{1}{16i} (2\pi i \cdot 6) = \frac{12\pi}{16} = \frac{3\pi}{4}$$

Exercise 5 (Graduate Only)

Evaluate the infinite series: $\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + 9}$.

Solution: Let $f(z) = \frac{\pi \cot(\pi z)}{z^2 + 9}$. We integrate over the large square contour C_N . The poles of $f(z)$ occur at $z = n$ (from $\sin(\pi z) = 0$) and at $z = \pm 3i$ (from $z^2 + 9 = 0$).

Residues at integers $z = n$:

$$\text{Res}(f, n) = \frac{\pi \cos(\pi z)}{\pi \cos(\pi z)(z^2 + 9)} \Big|_{z=n} = \frac{1}{n^2 + 9}$$

Residues at $z = \pm 3i$:

$$\text{Res}(f, 3i) = \frac{\pi \cot(\pi z)}{z + 3i} \Big|_{z=3i} = \frac{\pi \cot(3\pi i)}{6i} = \frac{\pi(-i \coth(3\pi))}{6i} = -\frac{\pi}{6} \coth(3\pi)$$

By the symmetry of the roots and the cotangent function, $\text{Res}(f, -3i) = -\frac{\pi}{6} \coth(3\pi)$ as well.

Applying the Residue Theorem:

$$\oint_{C_N} f(z) dz = 2\pi i \left[\sum_{n=-N}^N \frac{1}{n^2 + 9} - \frac{\pi}{6} \coth(3\pi) - \frac{\pi}{6} \coth(3\pi) \right]$$

Taking the limit as $N \rightarrow \infty$, the integral over C_N goes to 0:

$$0 = 2\pi i \left[\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + 9} - \frac{\pi}{3} \coth(3\pi) \right]$$

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + 9} = \frac{\pi}{3} \coth(3\pi)$$